

Youngmin Jo

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Education

Sogang University, Seoul, South Korea 3/2024 – 8/2027
B.S. Mechanical Engineering, Summa Cum Laude (expected)
Major GPA: 4.26/4.3, Overall GPA: 4.17/4.3

Research Experience

Undergraduate Researcher 12/2025 – present
Materials Design and Process Engineering Laboratory, **Sogang University**
Advisor: Prof. Heonjae Jeong

Thermodynamic Stability of Doped LLZO Solid-Electrolyte Interfaces 7/2026 – present

- Constructed interface structures of Al-, Ga-, Ta-, and Sr+Ta-doped LLZO solid electrolytes with the $\text{Li}_{13}\text{Si}_4$ anode and FeF_3 cathode for thermal-battery applications
- Evaluating the interfacial thermodynamic stability and Li-ion conductivity of each doped LLZO solid electrolyte at these interfaces

Fluoride-Ion Solid Electrolyte Synthesis Toward Inverse-Designed Compositions 8/2026 – present

- Benchmarking the group's first experimental workflow by synthesizing $\beta\text{-PbSnF}_4$ and validating its ionic conductivity against reference values before attempting inverse-design candidate compositions

Origin of Negative Photoconductivity in Amorphous ZnON 2/2026 – 8/2026

- Constructed amorphous ZnON (a-ZnON) by replicating the experimental sputtering process, using Gaussian, Packmol, and AIMD simulations with EDS-derived composition data
- Performed hybrid DFT (PBE0) calculations on a-ZnON to obtain accurate electronic structures
- Identified negative photoconductivity as originating from two coexisting trap channels — N 2p-derived hole localization and low-coordination Zn-derived electron localization — and quantified their trap energies at the valence/conduction band edges

In Preparation

W. Lee, W. Choi, H. Yoo, **Y. Jo**, M. Heo and H. Jeong, “Origin of Negative Photoconductivity in Amorphous ZnON” (title TBD)

Research Interests

First-principles and machine-learning approaches to thermal and charge transport in semiconductor materials
Synthesis and experimental validation of inversely designed materials

Technical Skills

Computational Materials Design: VASP, Gaussian, Pymatgen, ASE, Avogadro

Programming & Systems: Python, C, Linux/Bash, REST APIs

Engineering & Design: Arduino, MATLAB, AutoCAD, Autodesk Inventor, 3D Printing

Technical Projects

Monitoring and Logging Tools for Multi-Server VASP Calculations 7/2026 – 9/2026

- Developed a resource monitor reporting GPU utilizations and user job allocations on a shared Linux server
- Built job launchers that check GPU/core availability and flag conflicts before VASP runs, now used by the group
- Automated result logging and alerts by parsing VASP outputs for convergence, energy, and run settings,

then sending them to Google Sheets and Telegram via REST APIs across three compute servers

Computational Pre-Screening for Area-Selective ALD

3/2026 – 6/2026

Manufacturing Process — Top 5 report, selected for class presentation

- Investigated area-selective ALD as a bottom-up alternative to top-down etching, where edge placement errors limit 3D transistor patterning
- Proposed VASP-based pre-screening of substrate–inhibitor pairs to cut experimental time and cost, demonstrated with acetylacetone, which adsorbs on Al_2O_3 but not on SiO_2 (hydroxylated slabs)

Automated Robotic Arm Teleoperation System

3/2026 – 6/2026

- Developed a motion-driven robotic arm system capable of real-time teleoperation, aimed at automating repetitive lab experiments and manual labor tasks
- Streamlined the motor control and teleoperation pipeline by integrating the LeRobot package, replacing traditional hard-coding methods

Analysis for Aramid-Hybrid CFRP Fishing Rod

9/2025 – 12/2025

Fundamentals of Engineering Materials

- Showed in LS-DYNA that replacing the outermost CFRP ply of a tapered rod tube with aramid (AFRP) increases steel-sphere impact energy absorption by 130% and shifts failure from brittle fragmentation to ductile deformation
- Proposed the CFRP–AFRP hybrid layup, with a TiO_2 UV-protective coating for the aramid, to keep fishing rods reliable in harsh conditions: intense sunlight, salt exposure, and temperature swings

Weather-Aware Navigation Using Estimated Arrival Times

8/2025 – 12/2025

C Programming Basic

- Built a C navigation tool that estimates when a driver will reach each point on a route and shows the weather forecast for that point at that time, rather than a single snapshot at departure
- Integrated navigation, map, and Korea Meteorological Administration forecast APIs, matching estimated arrival times to hourly forecasts and sampling weather every 5 km along the route on an interactive map

Awards & Scholarships

President's Award & Dean's List with Highest Honors (Top 1%), Sogang University	Spring 2026, Fall 2025
Encouragement Prize, 11th DB GAPS Investment Competition, DB Financial Investment	10/2025
Best Team Award, DS Division, Samsung Electronics , South Korea	8/2025
Ilju Academy and Culture Foundation Scholarship, Taekwang Group, South Korea	8/2025 – 12/2027
Alumni Association Scholarship, Sogang University	3/2025

Leadership Experience

President (previously Vice President, Executive Committee Member)

3/2024 – 6/2027

MECHA, Mechanical Engineering Society, Sogang University

Direct engineering projects and manage a technical collaboration with Taejin Technology (HTC KOREA)

Team Leader, Samsung Shining Star 2025, Samsung Electronics, South Korea

8/2025

Led a project team through a training program focused on semiconductor foundry processes and the eight major fabrication steps, and presented the team's final project

Teaching & Outreach

English Tutor (Volunteer), Local Children's Center, South Korea

9/2025 – 6/2027

Deliver biweekly English lesson plans and academic mentorship to five middle school students

Technical Seminar Instructor, MECHA, Sogang University

3/2026, 5/2025

Designed and delivered a seminar on Arduino-based hardware prototyping, 3D modeling and printing

Global Volunteer (Exchange Participant), AIESEC in FHCMC, Vietnam

12/2024 – 2/2025

Taught English to secondary students and facilitated Korean language and cultural exchange sessions for university students over 210 hours of service